Cell Design Specification — t5_r1_flash

Doc No. VBF-T5R1FLASH-DS-01 · Design case: next-generation flagship-vehicle battery · Rev A (2026-08-25)

Requirement (task contract): energy density ≥ 500.94 Wh/kg · 4C fast charge without lithium plating · maximum temperature ≤ 60 °C (333.15 K).

1. Basic specification

Item	Value	Source
Electrochemical system	LNMO (LiNi0.5Mn1.5O4, 4.7 V high-voltage spinel) / graphite	base parameter set LNMO.json + Chen2020 negative defaults (runner-injected)
Nominal capacity	4.5 Ah (nominal, parameter set) / 8.26 Ah (simulated 1C, DFN)	LNMO.json `Nominal cell capacity [A.h]`; cell/r9_lnmo_r6_1c_dfn.json:capacity_ah
Voltage window	2.5 – 4.7 V	LNMO.json `Lower/Upper voltage cut-off [V]`
Cell dimensions	height × width × thickness = Not provided (area 0.1027 m^2, stack thickness 300 µm)	shell geometry not in parameter set; cell/r9_lnmo_r6_energy_dfn.json:thickness_m/area_m2
Electrolyte formulation	HiTrans-class carbonate/LiFSI-like transport: σ 6 S/m, t■ 0.70, D 1.0e-9 m^2/s (domain estimates, marked **estimate**, LiFSI-class literature; not yet true-MD-endorsed) + additive set FEC/VC/PES/DTD (funnel-passed, Stage 2)	params_lnmo_r6.json; funnel3 entry; values marked estimate in log
Cation transference number	0.70	params_lnmo_r6.json `Cation transference number`

2. Electrode and separator

Layer	Material	Thickness (µm)	Porosity	Active fraction	Source
Positive electrode	LNMO	98	0.30	0.665	params_lnmo_r6.json; LNMO.json active vol fraction
Negative electrode	Graphite	180	0.32	0.75	params_lnmo_r6.json; Chen2020 default active vol fraction
Separator	Not provided (default, ρ 397 kg/m^3)	9	0.55	—	params_lnmo_r6.json
Positive current collector	Al	8	—	—	params_lnmo_r6.json
Negative current collector	Cu	5	—	—	params_lnmo_r6.json
Particles	pos 2.5 µm / neg 1.5 µm	—	—	—	params_lnmo_r6.json

N/P ratio = (negative areal active capacity)/(positive areal active capacity)

= (m_neg·a_neg·q_neg)/(m_pos·a_pos·q_pos) = (0.2028×0.75×372)/(0.3018×0.665×146.6) = **1.92**

(q_neg 372 mAh/g graphite theoretical, q_pos 146.6 mAh/g LNMO theoretical — literature; audit round notes cited simplified estimates 1.78–2.14; deliverable value supersedes with explicit formula.)

3. Process design parameters

Parameter	Value	Formula / source
Positive areal density	302 g/m^2	thickness×(1–ε)×ρ = 98 µm×0.70×4400 kg/m^3 (calc-energy layer_kg_m2)

Parameter	Value	Formula / source
Negative areal density	203 g/m²	180 µm×0.68×1657 kg/m³ (calc-energy layer_kg_m2)
Positive compaction density	3.08 g/cm³	electrode density×(1−porosity) ÷1000
Negative compaction density	1.13 g/cm³	1657×0.68 ÷1000
Electrolyte fill amount	9.4 mL / 11.3 g	pore volume×electrolyte density×fill factor (1.0); electrolyte density 1.2 g/cm³ literature value, parameter set missing (annotated)
Formation recommendation	0.1C CC to 4.7 V, 25 °C, 2 cycles	design recommended value; production-line value requires tuning

4. Mass breakdown (per cell)

Component	Mass (g)	Source
Positive coating (LNMO composite)	31.00	calc-energy layer_kg_m2×area
Negative coating (graphite composite)	20.83	same
Positive CC (Al)	2.22	same
Negative CC (Cu)	4.60	same
Separator	0.17	same
Total (contract caliber, electrolyte excluded)	**58.81**	calc-energy mass_kg
Electrolyte (informational, 1.2 g/cm³)	11.3	pore volume calc (annotated literature density)
Total incl. electrolyte	70.1	above
Enclosure / tabs	Not modeled	beyond pure simulation boundary

Contract energy density excludes electrolyte per calc-energy contract (electrolyte_included=false); with electrolyte the cell would be 501 Wh/kg (informational).

5. Performance verification (conclusion-grade, DFN)

Metric	Value	Threshold (entry 0)	Determination
Energy density (1C, DFN)	597.9 Wh/kg	≥ 500.94	PASS PASS
Max temperature (4C charge @45 °C ambient, DFN)	327.55 K = 54.4 °C	≤ 333.15 K (≤60 °C)	PASS PASS
Lithium plating (4C, anode potential min)	+14.7 mV (> 0)	none (plated=false)	PASS PASS (no plating)
1C discharge capacity (DFN)	8.26 Ah	— (informational)	reported
Midpoint voltage	4.235 V	— (informational)	reported
DC resistance (calc-energy formula)	8.0 mΩ	— (informational)	reported
Volumetric energy density	1141 Wh/L	— (informational)	reported

All conclusion-grade values: cell/r9_lnmo_r6_energy_dfn.json, cell/r9_lnmo_r6_4c45_dfn.json (see DVPR-01 for per-row sources).

6. Design notes

- Baseline Chen2020 (NMC811/graphite): ED 400.3 Wh/kg, 4C anode min −0.438 V → confirmed below target; **escalated to Stage 2 materials** (plan_update1): LNMO 4.7 V high-voltage system + HiTrans-class transport + mass-cut architecture (10→8 µm Al, 6→5 µm Cu, 12→9 µm sep).

- ED chain: 492.8 → 520.2 → 539.5 → 566.8 → 622.6 → 612.6 → **603.6 (SPMe) / 597.9 (DFN)** — all ≥ 500.94 from R4 on; final architecture keeps 97 Wh/kg margin.

- Plating chain (anode min, same protocol, same 2.0 Ah charge cap): −0.128 → −0.041 → −0.005 → **+0.015 V** via N/P increase (122→180 µm negative), negative porosity 0.22→0.32, electrolyte D 4e-10→1e-9 m^2/s, σ 3.5→6 S/m, separator ε 0.45→0.55 (diagnosis in evaluate R6–R9 notes).

- T_max chain: 321–330 K across LNMO rounds, final 327.6 K at h=100 W/m^{2}K cooling (5.6 K margin).

- Transport values (σ 6 S/m, t■ 0.7, D 1e-9 m^2/s) are LiFSI-class **domain estimates** (marked estimate in audit; Stage 2 funnel passed the additive set; true DFT/MD endorsement skipped — real_compute=false, endorse entry records skip honestly).